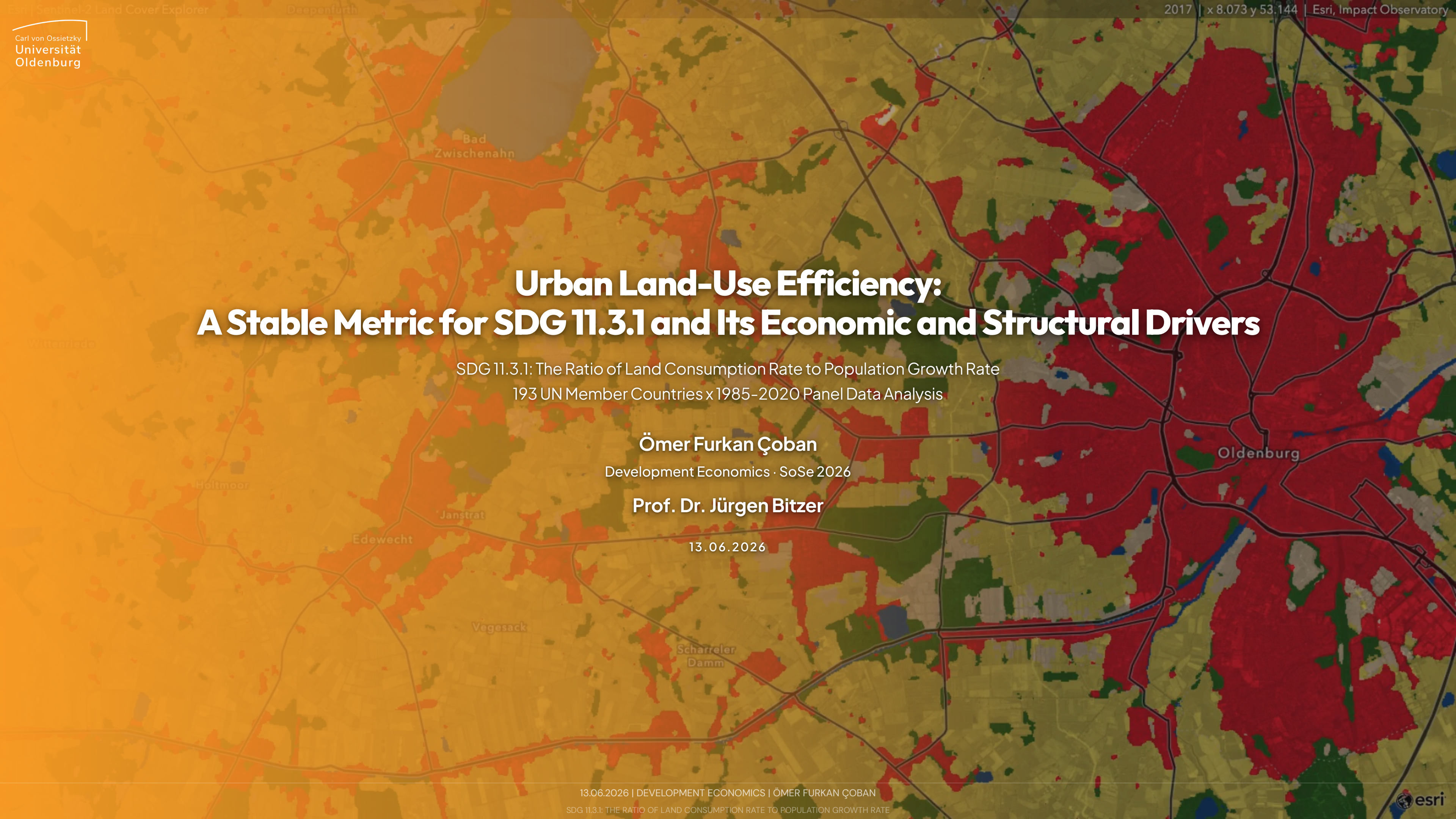




Urban Land-Use Efficiency: A Stable Metric for SDG 11.3.1 and Its Economic and Structural Drivers

SDG 11.3.1: The Ratio of Land Consumption Rate to Population Growth Rate
193 UN Member Countries x 1985–2020 Panel Data Analysis

Ömer Furkan Çoban
Development Economics · SoSe 2026

Prof. Dr. Jürgen Bitzer

13.06.2026

Introduction



SDG 11.3.1: Ratio of Land Consumption Rate to Population Growth Rate

A planet in motion

Current World Population	
8,303,260,254	
• Live projection synchronized with Worldometers demographic parameters	
TODAY	THIS YEAR
Births today 225,145	Births this year 68,048,247
Deaths today 106,089	Deaths this year 32,064,652
Population Growth today 119,056	Population Growth this year 35,983,595

Source: Live World Population Counter ([Worldometer 2024](#)).

- Every single year the world's built-up area grows by about the size of **the Netherlands**, as housing, roads and industry keep spreading outward ([Seto et al. 2012](#)).

But growth itself is not the problem

- Cities **must** grow to house more people; that is normal.
- What matters is **how much land each new resident uses**.
- If land grows **at the same pace as** population, the city stays **compact**.
- If land grows **much faster** than population, it **sprawls**.
- So the problem is **sprawl, not growth**.

Why It Matters?: Sprawl is a development problem

Public Finance:

- 📁 More land per person makes roads, water and transit cost far more per resident

Equity & Inclusion:

- 👤 Sprawl pushes poorer residents to the periphery and fuels informal settlements

Land & Climate:

- 🌿 Sprawl permanently builds over farmland, locks cities into car dependence, and drives high-emission growth

Measurement



SDG Indicator 11.3.1: How Is It Calculated?

Computing 11.3.1, step by step (UN-Habitat 2025)

STEP 0: DEFINE THE ANALYSIS AREA: THE *URBAN* AREA

- Computed **inside cities**, not over whole countries
- Urban area delineated by the **Degree of Urbanisation (DEGURBA)** → **GHS-SMOD** grid
- Built-up & population below measured **only within urban cells**

STEP 1: MEASURE BUILT-UP AREA AND POPULATION AT TWO POINTS IN TIME

Where:

- V_{past} : Urban built-up (m²), start of period
- $V_{present}$: Urban built-up (m²), end of period
- t : Years in the built-up window (= 5)
- Pop_t : Urban population, start of period
- Pop_{t+n} : Urban population, end of period
- y : Years in the population window (= 5)

STEP 2: LAND CONSUMPTION RATE (LCR)

$$LCR = \frac{V_{present} - V_{past}}{V_{past}} \times \frac{1}{t}$$

How fast is the city's physical footprint expanding per year? → **Arithmetic (linear) rate**

STEP 3: POPULATION GROWTH RATE (PGR)

$$PGR = \frac{\ln(Pop_{t+n}/Pop_t)}{y}$$

How fast is the population growing per year? → **Logarithmic (exponential) rate**

STEP 4: THE RATIO

$$LCRPGR = \frac{LCR}{PGR}$$

How to read the result



LCRPGR < 1: **Densification**

Population grows faster than built-up area. More people per km² of city over time.



LCRPGR = 1: **Balanced [SDG Target]**

Land consumption and population grow at equal rates. Sustainable expansion.



LCRPGR > 1: **Sprawl**

Built-up area expands faster than population. Rising infrastructure costs per resident, farmland loss, transit failure.

A Flawed Yardstick: Two Structural Problems

Problem 1: Mathematical asymmetry

LCR uses an **arithmetic** (linear) rate; PGR uses a **logarithmic** (exponential) rate.

$$\underbrace{\frac{V_{present} - V_{past}}{V_{past}} \times \frac{1}{t}}_{\text{arithmetic LCR}} \div \underbrace{\frac{\ln(Pop_{t+n}/Pop_t)}{y}}_{\text{logarithmic PGR}}$$

A city with 5% annual growth in **both** built area and population does **not** produce $LCRPGR = 1$, the two “5%s” are on different mathematical scales. Taking a ratio of two differently-defined rates is dimensionally inconsistent: cross-country comparisons are distorted by the formula, not by real urbanisation.

i Worked example: 5%/yr in both, over 5 years

Same growth factor $g = 1.05^5 = 1.276$ for built area **and** population:

$$LCR = \frac{g - 1}{5} = \frac{0.276}{5} = 5.53\%, \quad PGR = \frac{\ln g}{5} = \frac{0.244}{5} = 4.88\%$$
$$\Rightarrow \mathbf{LCRPGR = 1.13 \text{ (not 1)}}$$

Identical growth, yet the formula alone signals “sprawl” because $(g - 1) > \ln g$.

Problem 2: Division instability

When $PGR \approx 0$ (shrinking or stable populations), $LCRPGR$ diverges to $\pm\infty$.

Scenario	LCR	PGR	LCRPGR
Normal Growth	2%	1.5%	1.33
Shrinking Population	2%	-0.1%	-20
Stable Population	2%	0.001%	2,000

The same **2% land growth** reads as 1.33, -20, or 2,000 depending only on how close PGR is to zero. Because the denominator can vanish, the blow-up is **structural**: no winsorising or filtering repairs it, it only hides the symptom.

Built-up Area per Capita: The Metadata's Secondary Indicator

Recommended in the Official Metadata

The 11.3.1 metadata (UN-Habitat 2025) itself states LCRPGR is **hard to interpret** and proposes **two secondary indicators**, built from the *same inputs* (§4.f):

SECONDARY INDICATOR 1: BUILT-UP AREA PER CAPITA

$$BUP = \frac{UrBU_t}{Pop_t} \quad (\text{m}^2/\text{person})$$

→ Measures **land consumed per person**: the *state* at one point in time.

SECONDARY INDICATOR 2: TOTAL CHANGE IN BUILT-UP AREA

$$\text{Total change} = \frac{UrBU_{t+n} - UrBU_t}{UrBU_t} \quad (\%)$$

→ Measures **total urban expansion** over the period (omits population).

We analyse the **growth rate of built-up area per capita**, the dynamic form of secondary indicator 1 (log form following (Lu and Weng 2026)):

$$BpCR = \frac{\ln\left(\frac{UrBU_t/Pop_t}{UrBU_{t-z}/Pop_{t-z}}\right)}{z} = LCR_{\log} - PGR_{\log}$$

→ Measures the **annual trend in per-capita land use**: densifying (< 0) or sprawling (> 0).

Where:

- $UrBU_t$: urban built-up (m²) in year t
- Pop_t : urban population in year t
- z : period length in years (= 5)
- $t-z / t+n$: earlier / later year
- $LCR_{\log} = \ln(UrBU_t/UrBU_{t-z})/z$: annual log growth of built-up
- $PGR_{\log} = \ln(Pop_t/Pop_{t-z})/z$: annual log growth of population

From a Level to a Trend

① Built-up per capita(UN-Habitat 2025)

$$BUP = UrBU / Pop$$

The level: land per person, right now.



② Annual growth of BUP(Lu and Weng 2026)

$$BpCR = \Delta \ln(BUP)/z = LCR_{\log} - PGR_{\log}$$

The change: is it rising or falling?



③ BpCR (Our Dependent Variable)

- ▲ **BpCR > 0** Land > People **Sprawl**
- ◆ **BpCR = 0** Land = People **Balanced**
- ▼ **BpCR < 0** People > Land **Densification**

Literature



Literature Review: Drivers & Measurement

Drivers of Urban Expansion

A Meta-Analysis of Global Urban Land Expansion (Seto et al. 2011)

- **Data:** 326 remote-sensing studies, global, 1970–2000
- **Scope:** rates and drivers of global urban land expansion
- **Finding:** urban land grew **faster than population** almost everywhere; income drives it in most regions, population growth in India and Africa

Determinants of Urban Sprawl in European Cities (Oueslati et al. 2015)

- **Data:** panel of 282 European cities (1990, 2000, 2006)
- **Scope:** econometric determinants of sprawl (monocentric model)
- **Finding:** **rising income and population** expand the urban footprint; cities keep spreading even where population is flat

Measurement & SDG 11.3.1

Ratio of Land Consumption Rate to Population Growth Rate: Analysis of Different Formulations Applied to Mainland Portugal (Nicolau et al. 2019)

- **Data:** mainland Portugal, municipality level
- **Scope:** SDG 11.3.1: UN LCRPGR vs an LUE-derived formula
- **Finding:** the UN **LCRPGR formula is structurally unstable**; a log / LUE-based alternative is more robust

From Data Inconsistency to Reliable SDG 11.3.1 Monitoring: A Global Multi-Dataset Assessment of Land Use Efficiency (Lu and Weng 2026)

- **Data:** 6 built-up × 4 population datasets, 176 countries, 2000–2020
- **Scope:** reliability of LUE (11.3.1) across datasets
- **Finding:** LUE estimates **swing widely with dataset choice**; a **log-based** framework is needed for reliable monitoring

💡 Our Contribution

- **Metric:** The **log-based BpCR** (Nicolau, Lu)
- **Method:** Make it **dynamic** with a two-way FE + **GMM(Generalized Method of Moments)** panel of **193 UN Member Countries (1985–2020)** measuring **persistence** (path-dependence)

Data



- European Union, Pesaresi et al. (2024); Schiavina et al. (2023)

- Food and Agriculture Organization, FAO (2024)

- UN SNAAMA (2025); UN DESA WPP (2024); UN M49 / UN-CDP (2024)



Descriptive Statistics

Descriptive Statistics

Variable	N	Mean	SD	Min	Median	Max
BpCR	1,500	0.004	0.017	-0.114	0.004	0.110
LCRPGR (log)	1,500	1.299	17.752	-567.915	1.025	281.447
ln(GDP per capita)	1,500	8.394	1.518	4.795	8.299	12.139
Urban Pop. Share	1,500	0.675	0.168	0.122	0.704	1.000
ln(Urban Density)	1,500	7.577	0.575	5.719	7.602	9.618
Net Migration (% Pop)	1,500	-0.111	2.294	-70.990	-0.042	8.622

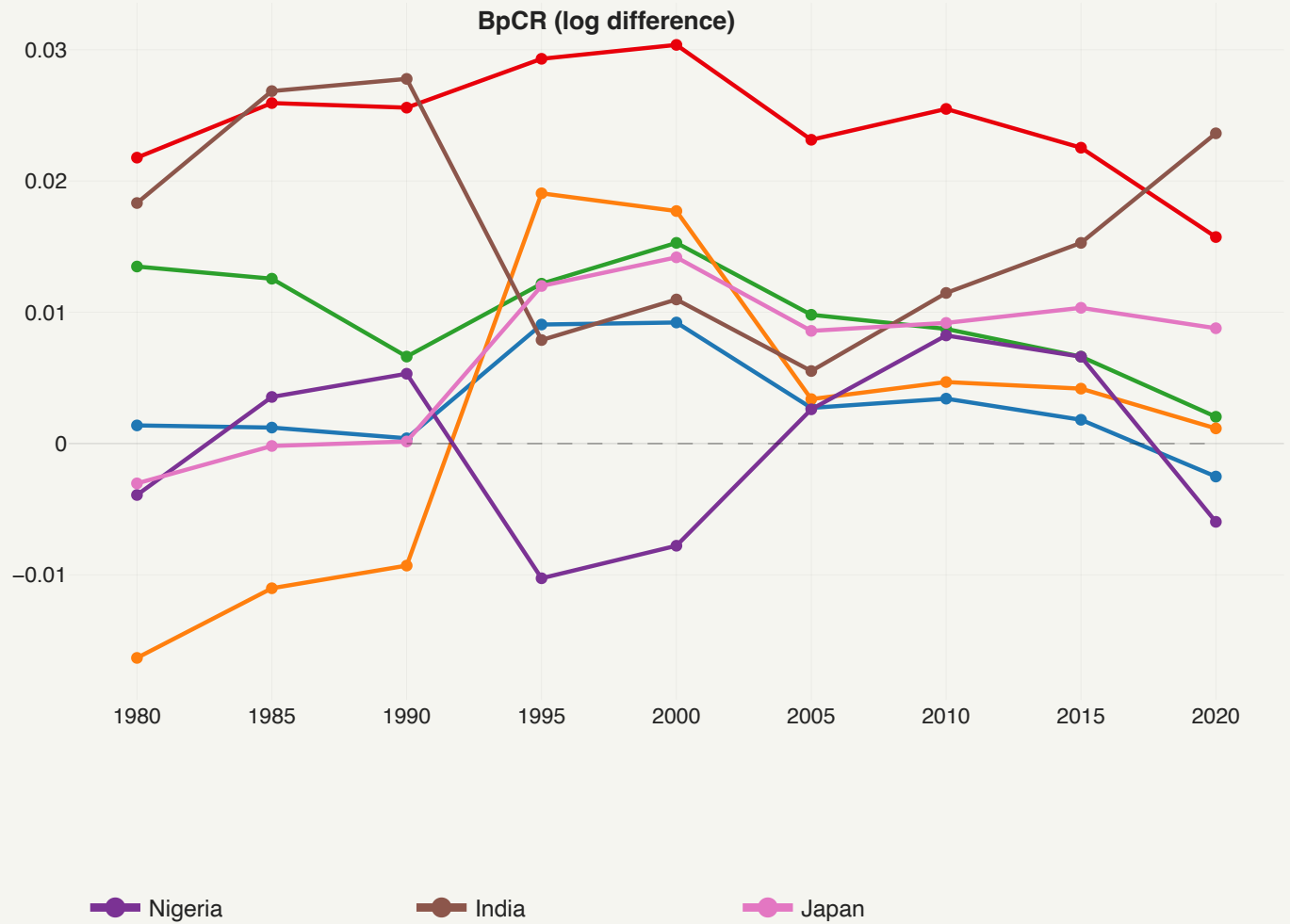
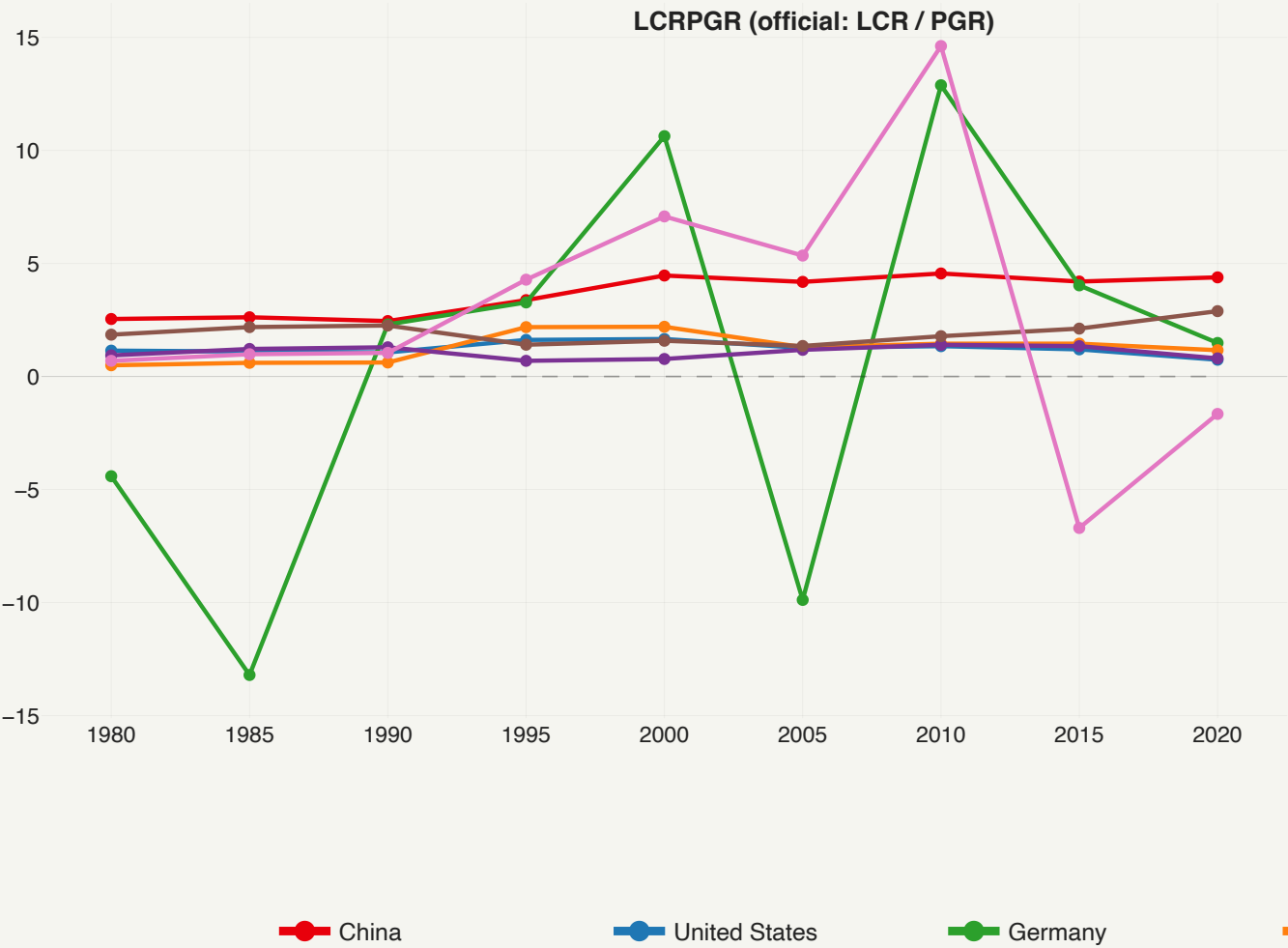
! Read the LCRPGR row

LCRPGR (log) already screams the problem: **SD ≈ 18** and a range from **-568 to +281**, vs **BpCR**'s SD ≈ 0.017 and range **[-0.11, +0.11]**. A $\sim 1,000\times$ wider spread: driven by LCRPGR's division instability when PGR ≈ 0 .

Country Trends: Regional Representatives

Urban Land Efficiency Trends: Regional Representatives

5-year periods, 1980-2020



i Key patterns

- **China:** strongest, most persistent per-capita sprawl (BpCR $\approx +0.02$ to $+0.03$), peaking around 2000 then easing to 0.016; LCRPGR drifts up to ~ 4.4 .
- **Germany:** trends toward balance (BpCR $0.015 \rightarrow 0.002$), yet its LCRPGR swings wildly (-13.2 to $+12.9$).
- **Japan:** BpCR negative in the 1980s, then a low, stable $\sim +0.01$, while its LCRPGR lurches from $+14.6$ to -6.7 .
- **Nigeria & Brazil:** oscillate around zero in BpCR (no persistent sprawl), with low LCRPGR (~ 1).
- **Takeaway:** BpCR gives a stable, interpretable per-capita signal; LCRPGR does not.

Empirical Analysis



Research Question & Hypotheses

Research question

What economic and structural factors drive within-country changes in per-capita urban land use over time, and how persistent are these trajectories?

 Measurement sub-question

Does the official SDG 11.3.1 indicator (**LCRPGR**) reliably capture this, or does the **choice of metric** change the conclusions?

Hypotheses

Hypothesis	Expected Sign
H1 BpCR measures efficiency better than LCRPGR	Higher Fit
H2 Higher urban density → less land/capita	–
H3 Higher income → less land/capita	–
H4 Net in-migration densifies cities	–
H5 Per-capita land use is path-dependent	+

Empirical Strategy

One specification, two dependent variables

$$\text{Metric}_{i,t} = \rho \text{Metric}_{i,t-1} + \beta_1 \ln(\text{UrbDens})_{i,t} + \beta_2 \text{UrbShare}_{i,t} + \beta_3 \ln(\text{GDPpc})_{i,t} + \beta_4 \text{NetMigr}_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t}$$

Two-stage plan

- ① Compare the metric **statically** (drop $\rho \text{Metric}_{i,t-1}$) to isolate **LCRPGR vs BpCR**
- ② **Add the lagged DV** for dynamics and correct its Nickell bias with **GMM** (*Final Model*)

The same model is estimated twice; only Y changes:

	Model 1	Model 2
DV	LCRPGR (UN)	BpCR (Ours)
FE	Country + Period	Country + Period
SE	Clustered (country)	Clustered (country)
N	1499	1499

- **Dynamic two-way FE panel:** country + period fixed effects (μ_i, δ_t)
- **Same RHS, two DVs** (LCRPGR vs BpCR): any divergence isolates the *metric*, not the data
- **Lagged DV** (ρY_{t-1}): captures inertia; the FE ρ is Nickell-biased downward, GMM-corrected (see Final Model)
- **Controls** (contemporaneous, at t): urban density, urban share, GDP pc, net migration
- **193 countries** $\times$ **8 five-year periods** (1985–2020); SE clustered by country

Why Each Variable Enters

Variables	Rationale
Lagged DV (ρ)	Captures inertia: once a country sprawls, it tends to keep sprawling
ln(Urban Density)	Urban form: denser cities consume less new land per resident
Urban Pop. Share	How urbanised the country already is; less rural land left to convert
ln(GDP Per Capita)	Tests whether richer countries use more or less built-up land per person
Net Migration	Demographic pressure: do newcomers densify cities or push outward expansion?
Country & Period FE	Absorb fixed geography and institutions, plus shocks shared in each period

Model 1: UN Approach: LCRPGR Regression

$$LCRPGR_{i,t} = \beta_1 \ln(UrbDens)_{i,t} + \beta_2 NetMigr_{i,t} + \beta_3 \ln(GDPpc)_{i,t} + \beta_4 UrbShare_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t}$$

Dependent variable: LCRPGR (log)				
(193 countries x 8 periods, 1985–2020)				
	Density	+ Net Migr.	+ GDP pc	+ Urban %
In(Urban Density)	-5.226	-5.226	-4.384	-2.608
	(3.613)	(3.614)	(3.220)	(1.947)
Net Migration (% Pop)		0.007	-0.005	-0.024
		(0.029)	(0.029)	(0.039)
In(GDP per capita)			2.662	2.740
			(1.617)	(1.682)
Urban Pop. Share				-14.932
				(16.141)
Observations	1499	1499	1499	1499
Within R²	0.001	0.001	0.003	0.003
Country FE	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes
* p < 0.1, ** p < 0.05, *** p < 0.01				
SE clustered at country level. DV: LCRPGR (log) = Land Consumption Rate ÷ Population Growth Rate (both log rates). Static two-way FE (no lagged DV); GHSL/SMOD + UN (GDP, net migration).				

Model 2: Our Approach: BpCR Regression

$$BpCR_{i,t} = \beta_1 \ln(UrbDens)_{i,t} + \beta_2 NetMigr_{i,t} + \beta_3 \ln(GDPpc)_{i,t} + \beta_4 UrbShare_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t}$$

Dependent variable: BpCR				
(193 countries x 8 periods, 1985–2020)				
	Density	+ Net Migr.	+ GDP pc	+ Urban %
In(Urban Density)	-0.0196***	-0.0196***	-0.0219***	-0.0218***
	(0.0070)	(0.0070)	(0.0070)	(0.0070)
Net Migration (% Pop)		-0.0008**	-0.0008**	-0.0008**
		(0.0003)	(0.0003)	(0.0003)
In(GDP per capita)			-0.0075***	-0.0075***
			(0.0028)	(0.0028)
Urban Pop. Share				-0.0014
				(0.0261)
Observations	1499	1499	1499	1499
Within R²	0.028	0.045	0.065	0.065
Country FE	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes
* p < 0.1, ** p < 0.05, *** p < 0.01				
SE clustered at country level. DV: BpCR = Δln(built-up per capita)/z ≡ LCR – PGR. Static two-way FE (no lagged DV); GHSL/SMOD + UN (GDP, net migration).				

Final Model: BpCR under Dynamic-Panel GMM

$$BpCR_{i,t} = \rho BpCR_{i,t-1} + \beta_1 \ln(UrbDens)_{i,t} + \beta_2 NetMigr_{i,t} + \beta_3 \ln(GDPpc)_{i,t} + \beta_4 UrbShare_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t}$$

The dynamic FE model is built up control-by-control (first five columns); the last two columns correct the lagged-DV **Nickell bias** with **Arellano-Bond** and **Blundell-Bond** GMM (instrumenting the lagged BpCR with its deeper lags).

Dependent variable: BpCR							
$\Delta \ln(\text{built-up area} \div \text{population}) / z \equiv \log\text{-LCR} - \log\text{-PGR}$							
(193 countries x 8 periods, 1985–2020)							
	Fixed Effects					GMM (Bias-Corrected)	
	Base	+ Density	+ Net Migr.	+ GDP	+ Urban %	Arellano-Bond	Blundell-Bond
BpCR[t-1]	+0.2535*** (0.0578)	+0.2346*** (0.0582)	+0.2346*** (0.0585)	+0.2286*** (0.0583)	+0.2287*** (0.0583)	+0.6057*** (0.1081)	+0.5908*** (0.0623)
ln(Urban Density)		-0.0147** (0.0064)	-0.0146** (0.0065)	-0.0169** (0.0065)	-0.0165** (0.0065)	-0.0523*** (0.0142)	-0.0021** (0.0010)
Net Migration (% Pop)			-0.0008** (0.0003)	-0.0008** (0.0003)	-0.0008** (0.0003)	-0.0005* (0.0003)	-0.0010*** (0.0002)
ln(GDP per capita)				-0.0068** (0.0028)	-0.0068** (0.0028)	-0.0138** (0.0060)	-0.0004 (0.0003)
Urban Pop. Share					-0.0031 (0.0225)	-0.0310 (0.0520)	+0.0079* (0.0045)
Observations	1,499	1,499	1,499	1,499	1,499	1,351	1,544
Within R²	0.076	0.091	0.108	0.125	0.125	-	-
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Instruments	-	-	-	-	-	15	21
Hansen (p)	-	-	-	-	-	0.44	0.14
AR(1) (p)	-	-	-	-	-	0.00	0.02
AR(2) (p)	-	-	-	-	-	0.71	0.47
* p<0.10, ** p<0.05, *** p<0.01. Standard errors clustered by country. Notes: 1) Cols 1–5: dynamic two-way FE (country + period FE), controls added stepwise. 2) AB = Arellano-Bond difference GMM; BB = Blundell-Bond system GMM, instrumenting the lagged BpCR with lags ≥ 2 (two-step, Windmeijer-robust SE). 3) Hansen J = over-identification test (p > 0.10 → valid instruments); AR(1)/AR(2) = first/second-order serial correlation (AR(1) significant by construction; AR(2) p > 0.10 → valid).							

 **BpCR[t-1]** **INERTIA**
ρ = +0.61*** ~60% of last period's BpCR persists; Bias-Corrected
ρ ≈ 3× FE → Strong path-dependence.
H5 ✓ Supported: Per-Capita Land Use is Path-Dependent (+)

 **ln(Urban Density)** **DENSIFICATION**
β₁ = -0.052*** **Compact-City Effect:** Denser cities use less new land per resident.
H2 ✓ Supported: Density → Less Land/Capita (-)

 **Net Migration** **DENSIFICATION**
β₂ = -0.0005* In-migrants pack into the **existing** stock faster than built-up grows.
H4 ✓ Supported: In-migration densifies (-)

 **ln(GDP per capita)** **DENSIFICATION**
β₃ = -0.014** Richer countries **densify, not sprawl** → **Growth ≠ Sprawl.**
H3 ✓ Supported: Higher income → less land/capita (-)

 **Urban Pop. Share** **NOT SIGNIFICANT**
β₄ = -0.031 (n.s.) The *level* of urbanisation adds nothing once the rest is controlled → urban **form** matters.
Control · No dedicated hypothesis

Conclusion



Conclusion

- **Sprawl is real and measurable.** In most world regions urban land grows faster than population: a policy-relevant inefficiency.
 - $\text{BpCR} > 0$ (sprawl) in **63%** of country-period observations; **67% of countries (129/193)** sprawl on average over time (Source: own calculations (GHSL/SMOD + UN data, 1985–2020))
 - Median $\text{BpCR} = +0.0041$ (built-up per capita rising); $\text{LCRPGR} > 1$ in **56%** of cases (Source: own calculations (GHSL/SMOD + UN data, 1985–2020))
- **The official LCRPGR is unreliable.** A raw ratio that explodes when population growth is near zero (division instability) and explains almost nothing within countries (within $R^2 \approx 0.003$).
- **Our log-based BpCR is the better metric.** Symmetric and numerically stable (following Nicolau and Lu), it fits about **20× better** (within $R^2 \approx 0.065$ static, ≈ 0.13 dynamic).
- **Robust drivers (Bias-Corrected GMM).** Denser cities densify (negative, highly significant); sprawl is **path-dependent** ($\rho \approx 0.6$); net migration densifies; and **higher income densifies too** (negative, significant).
- **We complement the literature, not contradict it.** Between countries, richer ones are more sprawled (the usual finding); but within a single country, growing richer leads to **densification** (building up, not out). Our country fixed effects capture this within-country effect that cross-country studies miss.
- **Methodological contribution.** A dynamic two-way FE + **GMM** panel of **193 UN member countries (1985–2020)** that corrects the Nickell bias and quantifies persistence.

References

- **Lu, W. & Weng, Q. (2026).** From data inconsistency to reliable SDG 11.3.1 monitoring: A global multi-dataset assessment of land use efficiency. *Land Use Policy*, 168, 108088.
- **Seto, K. C., et al. (2011).** A meta-analysis of global urban land expansion. *PLOS ONE*, 6(8).
- **Oueslati, W., Alvanides, S. & Garrod, G. (2015).** Determinants of urban sprawl in European cities. *Urban Studies*, 52(9), 1594–1614.
- **Nicolau, R., David, J., Caetano, M. & Pereira, J. M. C. (2019).** Ratio of Land Consumption Rate to Population Growth Rate: analysis of different formulations applied to mainland Portugal. *ISPRS Int. J. Geo-Information*, 8(1), 10.
- **UN-Habitat. (2025).** *SDG Indicator 11.3.1 Metadata: Ratio of Land Consumption Rate to Population Growth Rate*. SDG indicator metadata, harmonized template v1.1.
- **Pesaresi, M., Schiavina, M., Politis, P., Freire, S., Krasnodębska, K., Uhl, J. H., ... Kemper, T. (2024).** Advances on the Global Human Settlement Layer by joint assessment of Earth Observation and population survey data. *International Journal of Digital Earth*, 17(1). <https://doi.org/10.1080/17538947.2024.2390454>
- **GHS-BUILT-S (R2023A).** European Commission, Joint Research Centre (JRC). Product: GHS-BUILT-S; epoch 2020; 100 m; Mollweide. © European Union, 1995–2025: CC BY 4.0.
- **Schiavina, M., Melchiorri, M. & Pesaresi, M. (2023).** GHS-SMOD R2023A: GHS settlement layers, application of the Degree of Urbanisation methodology (stage I) to GHS-POP R2023A and GHS-BUILT-S R2023A, multitemporal (1975–2030). European Commission, JRC. doi:10.2905/AODF7A6F-49DE-46EA-9BDE-563437A6E2BA
- **FAO (2024).** The Global Administrative Unit Layers (GAUL) 2024. Food and Agriculture Organization of the United Nations: CC BY 4.0.
- **UN Statistics Division (2025).** National Accounts Main Aggregates Database (SNAAMA): per-capita GDP, constant 2020 US\$.
- **UN DESA, Population Division (2024).** World Population Prospects 2024: crude net-migration rate.
- **Worldometer (2024).** *World Population Clock: Current World Population*. <https://www.worldometers.info/world-population/> (real-time projection based on UN DESA WPP 2024).



Thank you for your attention!

Questions?



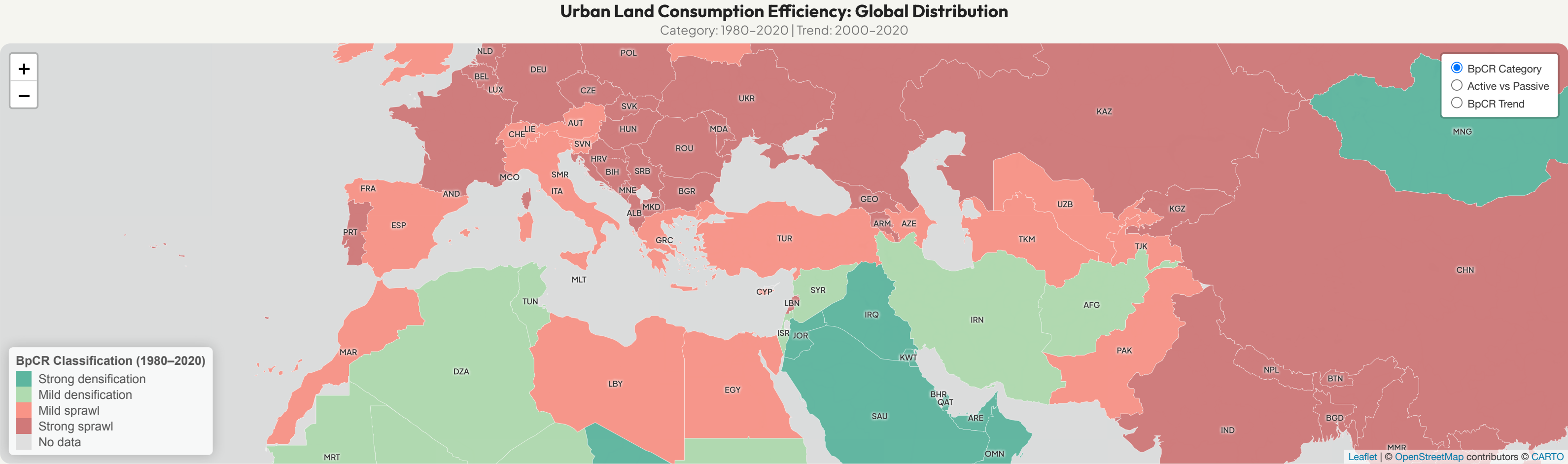
Appendix: Backup Slides



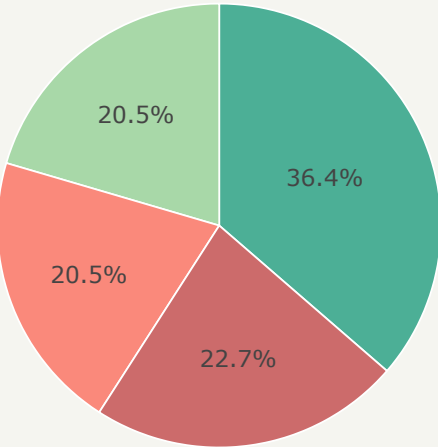
The Five Indicators at a Glance

Indicator	Formula (annualised)	Type	What it measures	Caveat
LCR · Land Consumption Rate	$\frac{UrBU_t - UrBU_{t-z}}{UrBU_{t-z}} \cdot \frac{1}{z}$ (arithmetic)	rate	how fast built-up area expands	ignores population
PGR · Population Growth Rate	$\ln(Pop_t / Pop_{t-z}) / z$ (log)	rate	how fast urban population grows	building block
LCRPGR · official 11.3.1	LCR / PGR	ratio	land growth relative to population growth	unstable: $\rightarrow \pm\infty$ as $PGR \rightarrow 0$; arithmetic $\div$ log
BUP · Built-up per capita (secondary 1)	$UrBU_t / Pop_t$	level	land consumed per person (the <i>state</i>)	a stock: mechanical as a regression DV
BpCR · our dependent variable	$\frac{\ln(BUP_t / BUP_{t-z})}{z} = LCR_{log} - PGR_{log}$	rate	the <i>trend</i> in per-capita land use (efficiency)	not named in metadata; growth of secondary 1

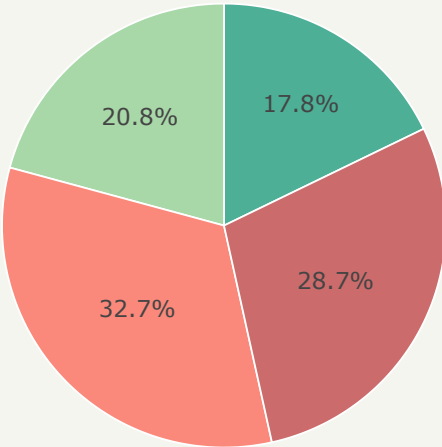
Metadata secondary indicator 2, Total change in built-up = $(UrBU_{t+n} - UrBU_t) / UrBU_t$, measures total expansion but, like LCR, omits population. $UrBU$ = urban built-up area, Pop = urban population, z = period length (5 yr).



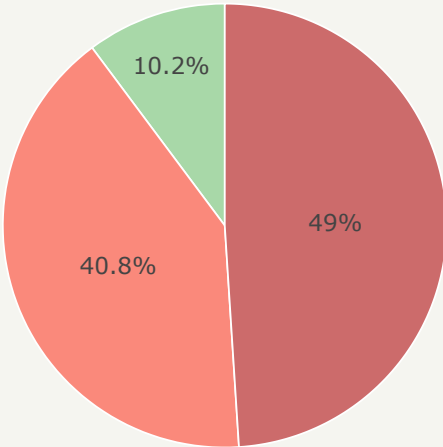
Distribution by Development Group (%)



LDC



Developing



Developed

Strong densification: $\text{BpCR} < -0.007$ Mild densification: $-0.007 \leq \text{BpCR} < 0$ Mild sprawl: $0 \leq \text{BpCR} < 0.009$ Strong sprawl: $\text{BpCR} \geq 0.009$

Country mean BpCR, 1980–2020. Thresholds: 0 (theory) and the median of each side.